

LAB 03

```
from sklearn.datasets import load_iris
from sklearn.decomposition import PCA
from sklearn.preprocessing import StandardScaler
import pandas as pd
import matplotlib.pyplot as plt

iris = load_iris()
features = iris.data
target = iris.target

scaler = StandardScaler()
features_standardized = scaler.fit_transform(features)

pca = PCA(n_components=2)
features_pca = pca.fit_transform(features_standardized)

pca_df = pd.DataFrame(data=features_pca, columns=["Principal Component 1", "Principal Component 2"])
pca_df["Target"] = target

plt.figure(figsize=(8, 6))

for label, color in zip(iris.target_names, ["red", "green", "blue"]):
    plt.scatter(
```

```
pca_df.loc[pca_df["Target"] == list(iris.target_names).index(label), "Principal Component 1"],  
pca_df.loc[pca_df["Target"] == list(iris.target_names).index(label), "Principal Component 2"],  
label=label,  
alpha=0.7  
)
```

```
plt.title("PCA on Iris Dataset (4 features to 2 features)", fontsize=14)  
plt.xlabel("Principal Component 1", fontsize=12)  
plt.ylabel("Principal Component 2", fontsize=12)  
plt.legend(title="Species")  
plt.grid()  
plt.show()
```

```
explained_variance = pca.explained_variance_ratio_
```

```
print("Explained Variance by each Principal Component:")  
print("Principal Component 1: ", explained_variance[0])  
print("Principal Component 2: ", explained_variance[1])  
print("Total Variance Retained: ", sum(explained_variance))
```